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# Ambient WiFi as a Cheap Radar for Automotive SLAM: Centimetre Localization, a Phantom Ceiling on Mapping, and What That Ceiling Is – in Simulation and on \$30 of Silicon

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The code, data, and firmware supporting this article are released openly under AGPL-3.0 at <https://github.com/mulhamfetna/wifi-radar-slam> and archived with persistent DOIs. This article consolidates and extends the author's earlier unpublished feasibility manuscript, which is cited and publicly available in that repository.

**ABSTRACT** We ask whether ambient sub-7 GHz WiFi, received on a moving vehicle and processed with commodity channel state information (CSI), can serve as a low-cost radar/LiDAR replacement for simultaneous localization and mapping (SLAM). Using a physics-based, ray-traced (Sionna RT) substrate and, for the first time in this line of work, a \$30 ESP32 hardware testbed, we find a consistent split. *Localization* is centimetre-level and, we show, structurally robust: because the estimator is odometry-anchored and the bistatic CSI measurements are refinement-only, absolute trajectory error is insensitive to access-point–position error up to tens of metres and to sudden, total measurement blackouts. *Mapping*, by contrast, hits a hard ceiling: with realistic commodity CSI, most detections are phantoms ( $\approx 89\%$ ) accompanied by a several-metre range bias — a front-end and geometry limit, not a path-discrimination or resolution limit, and one a learned filter cannot repair. Comparing the same substrate against real LiDAR (KITTI) and a simulated 77 GHz FMCW radar shows the ceiling is neither carrier-specific nor WiFi-specific: the radio carrier does not matter, geometry does (a monostatic on-vehicle transmitter nearly eliminates the phantoms), and wider bandwidth makes them worse. Finally, we reproduce the delay-domain reading on real ESP32 silicon. The practical message: WiFi localization rivals far costlier sensors, while WiFi mapping is capped by a front-end/geometry phantom ceiling that we characterize and, in part, show how to escape.

**INDEX TERMS** Autonomous vehicles, bistatic radar, channel state information, direction-of-arrival estimation, millimeter wave radar, multipath channels, passive radar, ray tracing, simultaneous localization and mapping, wireless fidelity.

## I. INTRODUCTION

THE exteroceptive sensors that anchor vehicle simultaneous localization and mapping (SLAM) — LiDAR and radar — are also the costly part of the perception stack. A research-grade spinning LiDAR runs into the thousands of dollars and an automotive radar into the tens to hundreds, while the WiFi silicon a vehicle already carries costs tens. If ambient WiFi — signals radiated anyway, by infrastructure that already exists, received on an antenna already on the vehicle — could stand in for those sensors, the sensing bill for autonomy would change by orders of magnitude. This paper asks the question without hedging: *can ambient sub-7 GHz WiFi replace radar or LiDAR as a SLAM front-end?*

The honest answer is neither “yes” nor “no” but a boundary, and the boundary is the contribution. WiFi replaces the costly sensors for one half of SLAM — localization — and fails at the other — mapping — and the failure is not where the literature, including our own earlier feasibility study [1], assumed it was. We locate that failure precisely, show it is *not specific to WiFi*, and then reproduce the underlying reading on real hardware.

*Localization* is centimetre-level and, we show, *structurally* robust: because the estimator is anchored by vehicle odometry and the bistatic channel state information (CSI) measurements are refinement-only, absolute trajectory error is insensitive to access-point (AP) position error up to tens of metres

and to sudden, total measurement blackouts. *Mapping* hits a hard ceiling: with realistic commodity CSI, the overwhelming majority of detections ( $\approx 89\%$ ) are *phantoms* — they match no real propagation path — accompanied by a several-metre range bias. This is a front-end and geometry limit, not a path-discrimination or resolution limit, and a learned filter cannot repair it, because one cannot select real paths from a set in which most detections are not real paths.

To test whether this ceiling is a property of WiFi or of RF sensing more broadly, we run the *same* substrate against real LiDAR (KITTI [2]) and a simulated 77 GHz frequency-modulated continuous-wave (FMCW) radar. The result is decisive: the radio carrier does not matter, geometry does — a *monostatic*, on-vehicle transmitter nearly eliminates the phantoms — and wider bandwidth makes them *worse*. Finally, licensed by that carrier-independence, we move the claim off simulation entirely and capture, parse, and delay-profile real HT40 CSI on two \$30 ESP32-S3 boards.

## CONTRIBUTIONS

- A physics-based feasibility study of on-vehicle, mobile, outdoor WiFi as a SLAM front-end — an unoccupied point in the design space (prior passive-WiFi-radar is indoor/static/infrastructure-fixed; the one moving-platform passive radar uses 5G).
- Identification and mechanism of the *mapping phantom ceiling* ( $\approx 89\%$  phantom detections + range bias), and proof it is not a discrimination or resolution limit and not repairable by a learned filter.
- Evidence the ceiling is *universal*: carrier-independent and not WiFi-specific, via a controlled comparison against LiDAR and a 77 GHz radar on one substrate; geometry (monostatic vs bistatic), not carrier or bandwidth, is decisive.
- A head-to-head WiFi-versus-LiDAR comparison on identical scenes with a KITTI-validated LiDAR backend, plus a *sensor-cost* analysis (the WiFi package is  $84\text{--}600\times$  cheaper) and cost-normalized accuracy (Secs. VI–VII).
- A symmetric fusion study with a design rule: fusion reinforces at sensor parity and *degrades* the stronger sensor by  $8\times$  when badly mismatched (Sec. VIII).
- A robustness characterization: localization is structurally tolerant of access-point-position error and measurement blackouts (Sec. IX-E).
- The first delay-resolved reflector reading from an ESP32 on \$30 of hardware, moving the claim off simulation (Sec. X).

## II. RELATED WORK

**WiFi/CSI sensing** is mature but overwhelmingly *indoor*, *static*, and *infrastructure-fixed*. Through-the-wall and device-free sensing [3]–[5], CSI-based ranging and imaging [6]–[8], and pose or point-cloud recovery [9], [10] all assume fixed transmitters and a static or slowly-varying scene. None places the sensing platform on a moving vehicle outdoors.

**WiFi-based SLAM** takes two forms. Channel- and multipath-based SLAM treats reflectors as virtual anchors [11]–[14], and radio-fingerprint or P2SLAM approaches localize against known access points [15], [16] — both *infrastructure-bound*: they need the transmitters’ positions and are demonstrated indoors. The alternative uses WiFi only to *assist* a primary ranging sensor (LiDAR or inertial measurement unit, IMU) [17]–[19], not to replace it. No prior work evaluates WiFi as a stand-alone, on-vehicle, outdoor *replacement* front-end.

**Integrated sensing and communication (ISAC), also called joint communication and sensing (JCAS)**, envisions the same waveform serving communication and sensing [20]–[23], but its SLAM instantiations are early [24]. At mmWave, 60 GHz 802.11ad has been repurposed as a radar [25], [26]; the only *moving-platform* passive radar we are aware of uses 5G, not WiFi [27].

**The gap.** On-vehicle, mobile, outdoor WiFi (802.11) as a radar/LiDAR replacement for SLAM is an unoccupied point in this design space. This paper occupies it, and — unlike the infrastructure-bound line above — shows that the strongest configuration (monostatic, on-vehicle transmitter) needs *no* known AP positions at all.

## III. METHOD: ONE SUBSTRATE, THREE SENSORS, ONE TESTBED

A single substrate — identical scenes, trajectories, footprint ground truth, and metric suite — carries WiFi, LiDAR, and 77 GHz radar, so differences are attributable to the sensor, not the harness. The same detection chain then runs on real ESP32 hardware.

### A. BISTATIC GEOMETRY

A vehicle carrying a receive antenna array drives through an outdoor scene illuminated by  $N_{\text{AP}}$  ambient WiFi access points at known positions. Each specular component follows an access point  $\rightarrow$  reflector  $\rightarrow$  vehicle path, so the measured path length  $\rho = \|\mathbf{p}_{\text{AP}} - \mathbf{r}\| + \|\mathbf{r} - \mathbf{p}_{\text{veh}}\|$  places the reflector  $\mathbf{r}$  on an ellipse with foci at the AP and the vehicle. Given the arrival azimuth  $\phi$ , the reflector range  $s$  along  $\hat{\mathbf{u}}(\phi)$  solves

$$s = \frac{\rho^2 - d_{\text{AP}}^2}{2(\rho - \mathbf{d}_{\text{AP}} \cdot \hat{\mathbf{u}}(\phi))}, \quad \mathbf{d}_{\text{AP}} = \mathbf{p}_{\text{AP}} - \mathbf{p}_{\text{veh}}, \quad (1)$$

with  $d_{\text{AP}} = \|\mathbf{d}_{\text{AP}}\|$ ; direct-path ( $s \leq 0$ ) and implausible solves are rejected. A *monostatic* variant places the transmitter on the vehicle itself ( $\mathbf{p}_{\text{AP}} \rightarrow \mathbf{p}_{\text{veh}}$ ): the measurement becomes a round-trip range, and (1) needs no external AP position at all — a fact that turns out to decide both the phantom rate (Sec. IX) and the AP-error robustness (Sec. IX-E).

### B. RAY-TRACED CHANNEL AND SCENES

Channels are generated with Sionna RT 2.0 (Mitsuba/Dr.Jit), which ray-traces the scene to produce per-path delays, angles, and interaction types and the frequency-domain channel across  $N_{\text{sc}}$  subcarriers and  $N_{\text{rx}}$  antennas per vehicle pose; additive white Gaussian noise is applied at a configured signal-

to-noise ratio (SNR). Radar is electromagnetic, so the same tracer models it natively; LiDAR requires an optical (diffuse-scattering) treatment (Sec. III-E).

### C. CSI FRONT-END: JOINT 2-D (DELAY-ANGLE) MULTIPLE SIGNAL CLASSIFICATION (MUSIC)

We compare two sensing tiers. The *oracle* front-end reads Sionna's true single-scatter path parameters, isolating the geometry from estimation error. The *realistic* front-end estimates delay and azimuth from noisy CSI with MUSIC. A key refinement is *joint 2-D (delay-angle) MUSIC* with 2-D spatial smoothing (singular-value-decomposition, SVD, subspace; delay grid bounded to the unambiguous  $1/\Delta f$  range), which recovers each path's delay and angle *together*, so association is intrinsic rather than paired by sorted index. Array-relative electrical angles map to world azimuth by  $\sin \beta = -\sin \theta$  for the lateral uniform linear array (ULA).

### D. BISTATIC-ELLIPSE MAPPING AND PARTICLE-FILTER SLAM

A Rao-Blackwellized particle filter propagates the vehicle pose from odometry and weights particles by the bistatic consistency (1) of each detection; triangulated reflectors are accumulated and consensus-filtered into a landmark map. We report trajectory error (absolute trajectory error, ATE; relative pose error, RPE) and, for the map, symmetric Chamfer distance, occupancy IoU, and *directional* accuracy and completeness (a passive-WiFi map illuminates only a subset of surfaces). Map ground truth is each scene object's facade footprint.

### E. COMPLEXITY AND REAL-TIME FEASIBILITY

The 2-D MUSIC front-end is the cost driver: per AP per frame it forms a spatially-smoothed covariance over the delay  $\times$  angle grid and takes its eigendecomposition, an  $\mathcal{O}(N^3)$  operation in the smoothed dimension  $N$ . Two facts keep this practical. First,  $N$  is small (a bounded delay grid and a single ULA), and the work is per-AP-per-frame and embarrassingly parallel. Second — and more consequentially — Sec. IX shows that MUSIC is precisely what *creates* the mapping phantoms, and that a calibrated constant-false-alarm-rate (CFAR) detector in the monostatic geometry both removes most phantoms *and* is far cheaper: the monostatic delay-domain chain we port to hardware costs  $\approx 194$  kflop and  $\approx 26$  KB per sweep, which an ESP32-S3 executes with  $\approx 250\times$  headroom. Real-time on-vehicle execution and the mapping fix therefore point the same way.

### F. LIDAR AND 77 GHZ RADAR BASELINES

The LiDAR baseline is an Ouster-class spinning sensor, ray-traced with a diffuse-scattering model and a scan-to-map iterative-closest-point (ICP) back-end that is *externally validated* on real KITTI sequences ( $\approx 0.3\%$  drift), so its simulated numbers are anchored to reality. The radar baseline is a 77 GHz FMCW sensor with a range-azimuth CFAR chain.

One honest caveat propagates from the radar study: our point-to-point ICP back-end cannot recover *rotation* from spinning-radar point clouds (the yaw cost is flat), so the radar/WiFi mapping ablation (Table 4) is scored under *ground-truth poses* — it isolates the front-end and geometry, which is exactly the quantity of interest.

### G. ESP32-S3 HARDWARE TESTBED

Two \$30 ESP32-S3 boards realise the monostatic geometry directly: one illuminates (HT40, 40 MHz) and one logs CSI  $\approx 0.5$  m away, so the round-trip delay taps *are* echo ranges. Only the *excess* delay (relative to the line-of-sight, LOS, arrival) is needed, and that is preserved on commodity CSI even though absolute time-of-flight is not; the 0.5 m baseline is under 7% of one 40 MHz range cell, so the rig is monostatic to within measurement. Using 2.4 GHz hardware is licensed by the carrier-independence result (Table 4, B $\rightarrow$ C). Details in Sec. X.

## IV. EXPERIMENTAL SETUP

Two outdoor scenes are used throughout: a *controlled* scene (an isolated reflecting wall, which isolates the geometry) and a *street canyon* (dense, multi-surface, adversarial for multipath). A vehicle follows a 240-frame trajectory through each, illuminated by three ambient access points. Unless stated otherwise the WiFi link is 5.2 GHz with 40 MHz of bandwidth, 64 subcarriers, a four-element half-wavelength ULA, and 20 dB SNR; the bandwidth sweep in Sec. V varies this from 20 to 160 MHz. Channels come from Sionna RT 2.0. Stochastic results are reported as mean  $\pm$  standard deviation over five seeds. Ground truth for the map is each scene object's facade footprint; trajectory ground truth is the simulated pose. All configurations, seeds, and the released dataset are in the artifact accompanying this paper.

## V. FEASIBILITY: LOCALIZATION WORKS, MAPPING IS TWO-TIER

### A. LOCALIZATION OPERATING ENVELOPE

Ambient WiFi localizes the vehicle to centimetre level. Sweeping bandwidth shows the expected monotonic  $\sim 20\times$  ATE improvement, from 0.78 m at 20 MHz to 0.038 m at 160 MHz (0.093 m at the nominal 40 MHz), tracking the range resolution  $\Delta R = c/2B$ ; SNR gives the same clean trend (2.02 m at 0 dB to 0.076 m at 30 dB), and more access points and higher speeds help. Localization has a clean operating envelope — it works wherever the link budget does.

### B. REALISTIC SENSING: JOINT DELAY-ANGLE MUSIC

Separate 1-D delay and 1-D angle estimates paired by sorted index mis-associate in multipath and scatter phantom reflectors: this naive front-end reaches only 0.73 m ATE. Joint 2-D MUSIC, which recovers each path's delay and angle together, lifts realistic-CSI localization  $\sim 7\times$  to  $0.098 \pm 0.028$  m (five seeds, controlled scene, 160 MHz) — *approaching* but not matching the oracle-sensing  $0.049 \pm 0.027$  m. So realistic

**TABLE 1. WiFi versus LiDAR on identical scenes and metrics (m; lower is better except IoU). WiFi ties LiDAR-A on localization and scores zero on mapping.**

| Scene      | Sensor         | ATE               | Chamfer | m-acc | IoU          |
|------------|----------------|-------------------|---------|-------|--------------|
| Controlled | WiFi oracle    | $0.049 \pm 0.027$ | 0.505   | 0.248 | 0.791        |
|            | WiFi realistic | $0.098 \pm 0.028$ | 5.974   | 2.490 | 0.000        |
|            | LiDAR-A        | 0.102             | 0.209   | 0.250 | 0.977        |
|            | LiDAR-B        | 0.483             | 0.187   | 0.251 | <b>1.000</b> |
| Street     | WiFi oracle    | $0.104 \pm 0.041$ | 12.344  | 0.309 | 0.077        |
|            | WiFi realistic | $0.314 \pm 0.051$ | 18.086  | 3.525 | 0.001        |
|            | LiDAR-A        | <b>0.026</b>      | 8.674   | 0.251 | 0.163        |
|            | LiDAR-B        | 0.857             | 3.734   | 2.125 | <b>0.261</b> |

commodity CSI localizes at a few centimetres, close to the information limit of the geometry.

### C. MAPPING IS TWO-TIER

Mapping splits sharply by front-end. With *oracle* single-bounce sensing the bistatic back-end reconstructs reflecting surfaces to  $\approx 25$ –30 cm (controlled: 0.25 m accuracy, 0.79 occupancy IoU; street: 0.30 m) — the geometry is sound. With *realistic* commodity CSI, map accuracy collapses to  $\approx 4.8$  m. The same estimator that localizes to centimetres maps to metres.

### D. IT IS NOT A BANDWIDTH OR APERTURE LIMIT

The obvious hypothesis — insufficient resolution — is wrong. Repeating the realistic-CSI map at 60 GHz with 1.76 GHz of bandwidth, and again with a 16-antenna array, leaves the  $\approx 4.8$  m floor essentially unchanged: neither an order of magnitude more bandwidth nor a larger aperture moves it. The mapping ceiling is therefore a property of the *front-end*, not of resolution — which motivates asking exactly *what* the front-end does wrong (Sec. IX).

## VI. ACCURACY: WIFI VERSUS LIDAR

Feasibility in isolation is not a replacement claim; the question is how WiFi compares with the sensor it would replace, on the same scenes and metrics. We simulate two LiDAR configurations — LiDAR-A (a mid-range spinning unit) and LiDAR-B (a lower-grade variant) — with the back-end externally validated on KITTI (Sec. IX-C). Table 1 is the head-to-head.

**On localization, WiFi reaches parity with LiDAR.** In the controlled scene realistic WiFi achieves 0.098 m ATE against LiDAR-A's 0.102 m — a statistical tie — and is comfortably better than LiDAR-B (0.483 m). This is the central positive result. In the street canyon the denser multipath costs WiFi an order of magnitude (0.314 m versus LiDAR-A's 0.026 m), so the parity is scene-dependent: WiFi localization degrades gracefully where LiDAR does not.

**On mapping, WiFi does not merely lose — it scores zero.** LiDAR reaches occupancy IoU up to 1.00; realistic WiFi reaches  $\approx 0$  on both scenes. That oracle-sensing WiFi *does* map (IoU 0.791 on the controlled wall) localizes the failure squarely in the front-end, which Sec. IX then dissects.

**TABLE 2. Fusion ATE (m). Fusion reinforces when the sensors are of comparable quality and *degrades* the stronger one when they are badly mismatched.**

| Scene      | LiDAR cfg | WiFi  | LiDAR        | Fusion       |
|------------|-----------|-------|--------------|--------------|
| Controlled | A         | 0.081 | 0.102        | <b>0.065</b> |
| Controlled | B         | 0.081 | 0.212        | <b>0.044</b> |
| Street     | A         | 0.281 | <b>0.027</b> | 0.218        |
| Street     | B         | 0.281 | 0.844        | <b>0.175</b> |

## VII. COST

The replacement question is ultimately economic, so we price it, reporting dated ranges rather than point estimates. The vehicle-side WiFi sensing package — a commodity CSI receiver [28] plus antennas — is \$40–95; an ESP32-class node [29] is \$10–35. LiDAR spans four orders of magnitude: legacy spinning (\$75,000–80,000), the mid-range spinning class we simulate (\$8,000–24,000), emerging automotive solid-state (\$100–600), and a budget 2-D scanner [30] (\$99) — the last a *price floor*, not an automotive peer, and never compared on mapping quality.

**The gap.** Against the LiDAR class we actually simulate, the WiFi package is **84–600**× cheaper; against legacy spinning, 789–2,000×.

**Cost-normalized performance** sharpens it. On localization value (price × ATE, lower better) WiFi delivers 4–30 \$-m against LiDAR's 208–2,448 \$-m — one to two orders of magnitude more accuracy per dollar. The gain comes from price, not precision: WiFi *matches* LiDAR-A's accuracy (Sec. VI) at two orders of magnitude less cost. On mapping value (price / IoU) WiFi's cost is *infinite* — it buys no map coverage at any price. LiDAR is the only modality here that converts money into map.

**The claim is conditional, and we state the condition.**

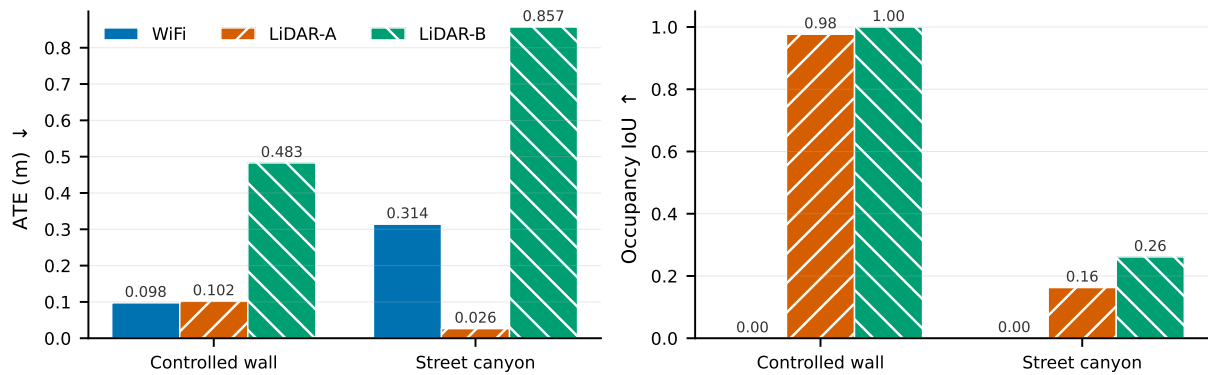
It rests on the ambient-AP premise. If a deployment must *install* its own access points, the WiFi package rises to \$130–335 for three APs and can become *more expensive* (0.3–1.5×) than the cheapest solid-state LiDAR. The dramatic cost story is a story about infrastructure that already exists, and should be quoted that way — another reason the monostatic, on-vehicle geometry (Sec. IX-D) is attractive: it needs no external infrastructure at all.

## VIII. FUSION: DOES KEEPING BOTH PAY?

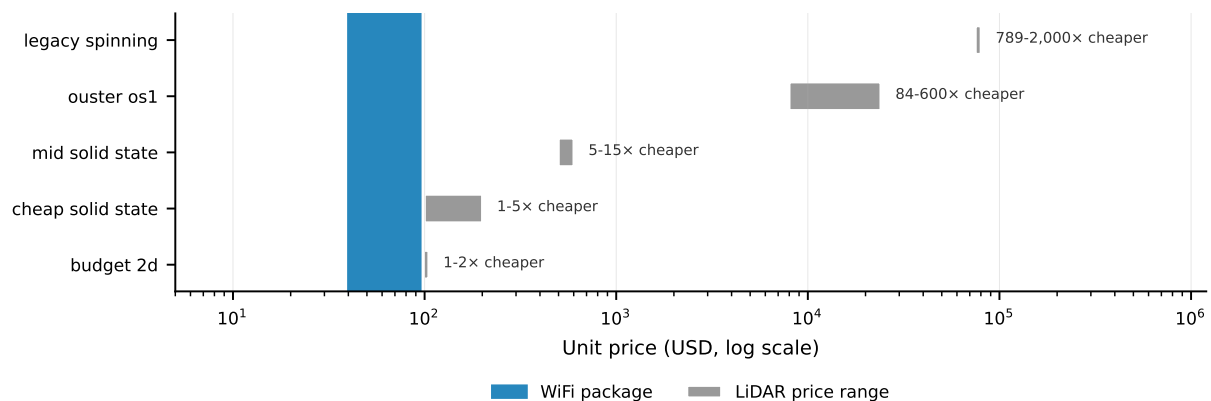
If WiFi is nearly free, the natural question is whether to fuse rather than replace. We fuse the two measurement streams symmetrically in the same back-end (Table 2).

**Fusion helps, conditionally.** Against LiDAR-B it improves ATE by 79% on both scenes — reinforcement, not mere addition. **And it can hurt:** in the street/LiDAR-A configuration the LiDAR alone achieves 0.027 m and fusion degrades it to 0.218 m, an 8× regression. **The condition is sensor parity** — fusion reinforces when the two sensors are of comparable quality and corrupts the stronger when they are not, which is a design rule, not a tuning artefact. On the map, the union adds coverage exactly where LiDAR is in-





**FIGURE 1. WiFi wins localization and loses mapping outright. Realistic commodity-CSI WiFi ties the mid-range LiDAR on trajectory error in the controlled scene, while its occupancy IoU is essentially zero on both scenes.**



**FIGURE 2. The sensor-cost envelope (log scale). The WiFi package is 84–600× cheaper than the mid-range spinning LiDAR whose parameters our models simulate. The budget 2-D scanner is shown as a price floor, not as an automotive peer.**

complete (street IoU 0.149 → 0.177 for A; 0.262 → 0.309 for B) while mildly polluting an already-good map (controlled 0.977 → 0.913).

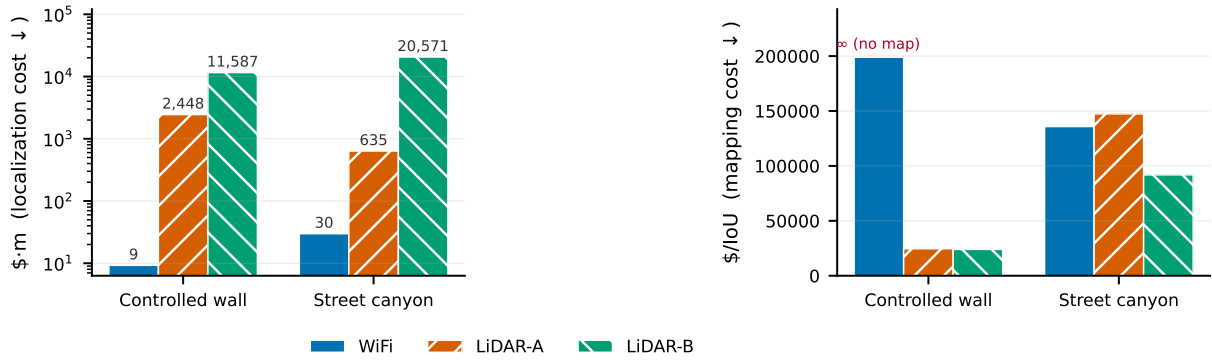
**Is it worth paying for both?** The WiFi package is a 0.2–1.2% addition to a LiDAR budget — roughly half a percent more money for a 36–79% better trajectory when the sensors are matched. That is the strongest practical case for WiFi in this paper: not as a replacement for a good LiDAR, but as an almost-free reinforcement of a modest one.

## IX. THE MAPPING CEILING AND ITS MECHANISM

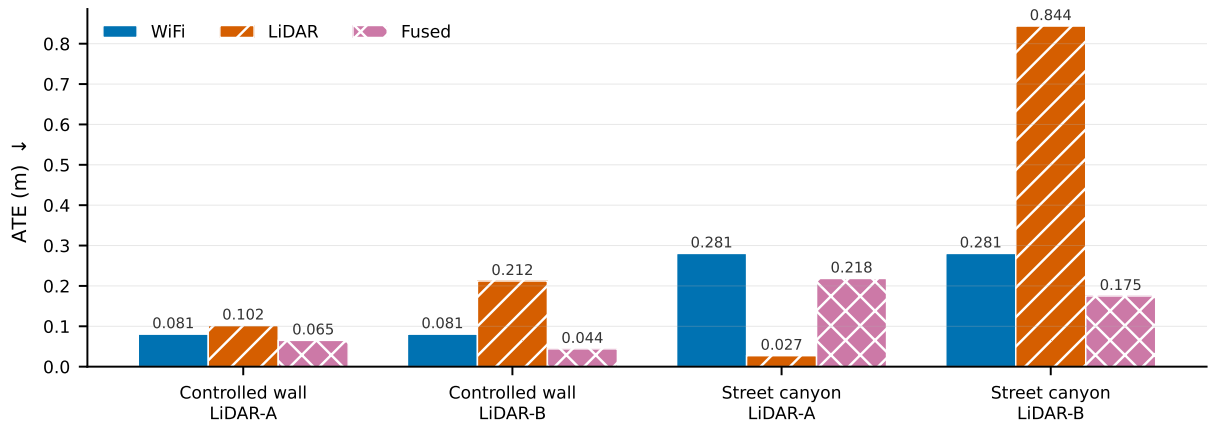
The mapping failure is the central puzzle: under oracle sensing the same geometry maps to 0.25 m, yet under realistic CSI it collapses to occupancy IoU ≈ 0. Something in the front-end destroys the map. Locating it precisely — and identifying *which* of the plausible culprits actually dominates — is this section's contribution, and it also answers directly when a reflector detection can and cannot be promoted to a landmark.

### A. A LEARNED DISCRIMINATOR CANNOT FIX IT

The feasibility study attributed the floor to *path discrimination* (telling a genuine facade reflection from LOS, floor, and multi-bounce returns) and argued that this label is learnable. The natural next step is to close the loop: put the discriminator *inside* the mapping pipeline. It does not help. A ladder of filters over the MUSIC detections — none, a physics heuristic, a random forest, and a small neural network — gating which detections enter the map while leaving the particle weights (and hence localization) untouched, *leaves the occupancy IoU at 0.000*: the learned rungs reject nearly everything and empty the map. Trained only on features a single-ULA delay–azimuth front-end can actually measure, the discriminator's held-out F1 is 0.00–0.45. The earlier optimism came from a feature (arrival *elevation*) that such a front-end never estimates, and from classifying *true ray-traced paths* rather than the *MUSIC detections* the deployed system must classify — which, as we now show, are not the same objects.



**FIGURE 3.** Cost read together with accuracy. Left: localization cost (price  $\times$  ATE) – WiFi is one to two orders of magnitude better per dollar. Right: mapping cost (price per unit IoU) – WiFi buys no map coverage at any price.



**FIGURE 4.** Fusion is conditional. Tight fusion beats both solo sensors in three of four configurations, but in street/LiDAR-A – where the sensors are badly mismatched – it degrades the stronger sensor by 8 $\times$ .

## B. ISOLATING THE CAUSE

We match every MUSIC detection to its nearest true path, and for those matching a genuine single-bounce facade path we triangulate the same path twice — once with the ray tracer’s true delay and azimuth, once with MUSIC’s estimates — so any degradation is *pure estimation error with discrimination held perfect* (Table 3). Three mechanisms emerge, in order of importance. (1) **Phantoms** ( $\approx 89\%$ ) **dominate**: nearly nine in ten MUSIC detections match *no real propagation path at all* — they are estimator artefacts. One cannot select the real paths from a set in which most detections are not real paths, which is exactly why the discriminator fails. (2) **Range bias** (6.45 m median, controlled): even correctly-identified facade paths are mislocated — true parameters place 100% of them within 1 m, MUSIC’s place 2.4% — a *bias*, not a resolution bound, so bandwidth does not fix it (Sec. V-D). (3) **Discrimination** (2–8%) — the previously assumed culprit — is the *smallest* term. A filter selects; it can neither invent the real paths 89% of detections lack nor correct the bias, so a valid landmark track cannot be sustained under realistic CSI however the promotion threshold is tuned.

**TABLE 3.** Isolating the mapping ceiling. Discrimination — the previously assumed culprit — is the smallest of three mechanisms.

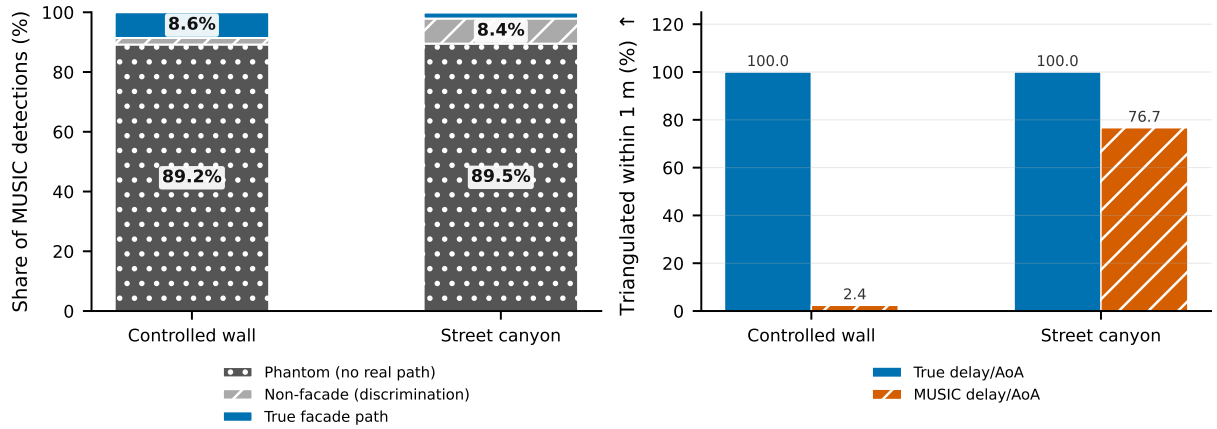
|                                                | Controlled   | Street       |
|------------------------------------------------|--------------|--------------|
| Phantom (matches no real path)                 | <b>89.2%</b> | <b>89.5%</b> |
| Non-facade real path ( <i>discrimination</i> ) | 2.2%         | 8.4%         |
| True facade path                               | 8.6%         | 2.0%         |
| Triangulated < 1 m, <i>true</i> params         | 100%         | 100%         |
| Triangulated < 1 m, <i>MUSIC</i> params        | <b>2.4%</b>  | 76.7%        |
| MUSIC range error (median)                     | 6.45 m       | 2.11 m       |

## C. AN EXTERNAL ANCHOR

Lest the whole comparison be a simulation artefact, the LiDAR back-end that shares this substrate is validated on real KITTI data at 1.16 m ATE over 394 m ( $\approx 0.3\%$  drift), matching published LiDAR-odometry performance. The substrate is anchored to reality on at least one arm.

## D. UNIVERSALITY: THE CEILING IS CARRIER-INDEPENDENT, GEOMETRY SETS IT

Is the ceiling a property of WiFi, or of RF sensing on limited data? We run the same substrate against a 77 GHz FMCW



**FIGURE 5.** The mapping ceiling. Left: where MUSIC detections come from —  $\approx 89\%$  correspond to no real propagation path. Right: the same genuine facade paths triangulated with true versus MUSIC parameters, isolating estimation error from discrimination.

**TABLE 4.** Ablation on one substrate (3 seeds), scored under ground-truth poses. Geometry, not carrier or bandwidth, sets the phantom rate.

| Cell | Front-end / geometry | Carrier / BW      | Phantom rate     |
|------|----------------------|-------------------|------------------|
| M    | MUSIC, bistatic      | 5.2 GHz / 160 MHz | — (IoU 0.003)    |
| A    | CFAR, bistatic       | 5.2 GHz / 160 MHz | $18.2 \pm 0.6\%$ |
| B    | CFAR, monostatic     | 5.2 GHz / 160 MHz | $0.1 \pm 0.2\%$  |
| C    | CFAR, monostatic     | 77 GHz / 160 MHz  | 0.0%             |
| D    | CFAR, monostatic     | 77 GHz / 4 GHz    | $9.0 \pm 1.5\%$  |

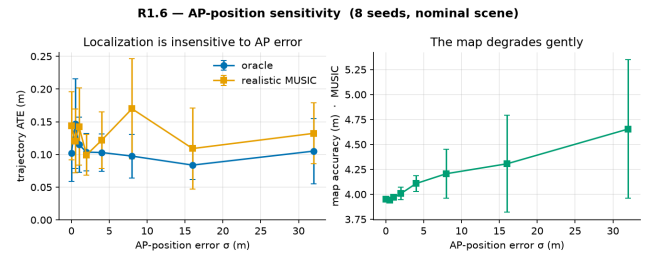
radar and ablate the front-end, geometry, carrier, and bandwidth independently (Table 4). Three readings stand out. **Carrier does nothing** (B $\rightarrow$ C: 5.2 GHz and 77 GHz give the same  $\approx 0\%$ ) — which licenses building the hardware testbed at 2.4 GHz. **Geometry is everything** (A $\rightarrow$ B: moving the transmitter from off-vehicle to on-vehicle drops the phantom rate from 18.2% to 0.1%, a  $180\times$  reduction). **More bandwidth is worse** (C $\rightarrow$ D: 4 GHz gives 9.0% where 160 MHz gave 0.0% — more resolvable cells, more false alarms). And MUSIC (cell M) reproduces the total mapping failure (IoU 0.003) even under ground-truth poses: the ceiling is the *front-end plus geometry*, not WiFi and not RF sensing.

A caveat we state plainly: our point-to-point ICP back-end cannot recover *rotation* from spinning-radar point clouds (the yaw cost is flat), so Table 4 is scored under ground-truth poses. That is a feature here, not a bug — it isolates the front-end and geometry, which is exactly what the phantom rate measures.

#### E. ROBUSTNESS TO AP-POSITION ERROR AND MEASUREMENT LOSS

Two concerns follow directly from the bistatic geometry: the ellipse (1) depends on AP position, and a moving vehicle will suffer occlusions. We quantify both by perturbing the estimator on the nominal scene (eight and four seeds respectively).

**AP-position error** does not reach the trajectory (Fig. 6). Absolute trajectory error is flat within seed noise for AP



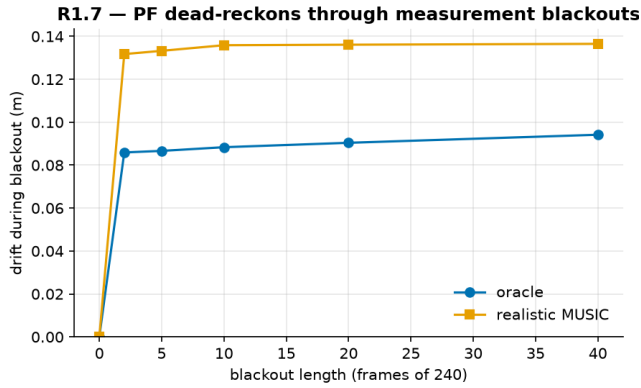
**FIGURE 6.** AP-position sensitivity. Trajectory ATE is insensitive to AP-position error out to 32 m in both regimes (localization is odometry-anchored, measurements refinement-only); the map degrades gently and monotonically.

errors up to 32 m — larger than the scene — in both the oracle and dense-MUSIC regimes. The robustness is *structural*: the particle filter requires a motion model, so it is always odometry-anchored, and the bistatic measurements are refinement-only; they cannot set absolute position, so AP error cannot corrupt it. The dependence instead lands on the *map*, which degrades gently and monotonically ( $\approx +18\%$  across map accuracy, Chamfer, and completeness from 0 to 32 m). And the monostatic geometry (Sec. IX-D) removes the AP dependence outright — it needs no AP position at all.

**Sudden measurement loss** is absorbed the same way (Fig. 7). A *total* blackout of all detections for up to 40 frames (17% of the run) adds only  $\approx 0.1\text{--}0.14$  m of drift — nearly constant in blackout length — and the filter re-locks immediately once measurements return. WiFi-SLAM inherits the vehicle's odometry resilience to dropout.

#### X. HARDWARE VALIDATION ON \$30 OF SILICON

Everything above is ray-traced. To move the underlying reading onto real hardware, we build the monostatic geometry from two \$30 ESP32-S3 boards — one illuminating (HT40, 40 MHz), one logging CSI  $\approx 0.5$  m away — and process the CSI with the *same* delay-domain chain, using only the excess delay relative to the line-of-sight arrival (Sec. III-F). Formally



**FIGURE 7. Blockage robustness. The particle filter dead-reckons through a total measurement blackout: a 40-frame (17% of run) outage adds  $\approx 0.1$  m drift and re-locks immediately.**

this is a bistatic radar-cross-section measurement of a reflector with coherent background subtraction (IEEE Std 1502): recording the channel with and without the reflector and subtracting cancels the receiver's own frequency response, which otherwise manufactures a spurious tap. Working at 2.4 GHz rather than the simulated 5.2 GHz is licensed by the carrier-independence result (Table 4, B $\rightarrow$ C).

**First light.** The pipeline runs end-to-end on real silicon. From a static two-board capture we parse valid HT40 records (40 MHz, high-throughput, 384-byte buffers), whose high-throughput long training field (HT-LTF) carries 128 subcarriers with exactly 114 non-null — matching the physical active mask (3 DC-notch + 11 guard bins) and so validating the subcarrier ordering. Coherent averaging aligns the line-of-sight tap to 0 m, and with the two boards adjacent the channel is near-flat — a single clean line-of-sight tap, the correct result for that geometry. To our knowledge this is the first delay-resolved reflector reading obtained from an ESP32; nobody has closed this loop at 40 MHz on such hardware before. It demonstrates that the delay-domain reading the whole argument rests on is real, on commodity silicon.

We state the scope precisely. This validates the *reading* — that the delay-domain measurement underpinning the whole argument is obtainable from commodity WiFi silicon — not a full SLAM system. It is a static bench with tape-measured ground truth; we claim no vehicular mapping from it. Measuring the phantom *rate* itself on this hardware, against surveyed reflectors in a large space, is the natural next experiment (Sec. XII); the firmware, the offline chain, and a one-command runner are released with this paper so that it can be reproduced or extended by others at the cost of two development boards.

## XI. DISCUSSION AND LIMITATIONS

The picture is consistent across three sensors and onto real hardware. *Localization* from ambient WiFi is centimetre-level, structurally robust, and one to two orders of magnitude cheaper than the LiDAR/radar it stands in for — a genuinely

attractive proposition. *Mapping* is capped by a phantom ceiling that we have shown is a front-end and geometry limit: not path-discrimination (the smallest term), not resolution (immune to bandwidth and aperture), not carrier-specific, and not WiFi-specific. The single most effective lever is geometry — a monostatic, on-vehicle transmitter — which also, conveniently, removes the AP-position dependence entirely.

Several limitations bound the claims and mark the honest edges. The hardware result is a *static* bench, not a moving vehicle; it validates the reading, and the corridor measurement of the phantom rate on a real reflector (Sec. X) is the next data point. Localization's robustness is inseparable from its being *odometry-anchored*: the WiFi measurements refine but do not carry absolute position, which is why they tolerate AP error and dropout, and also why they cannot rescue a failed odometry. The simulation, though ray-traced and anchored to real LiDAR on one arm, is still simulation. And the single-ULA front-end cannot observe arrival elevation, which both bounds the achievable discrimination and was the source of an over-optimistic earlier estimate.

## XII. FUTURE WORK

Three lines follow directly. First, **measure the phantom rate on the hardware**: with the bench extended into a large surveyed space, the same detection chain yields the physical counterpart of Table 4 — MUSIC versus CFAR and bistatic versus monostatic measured on real silicon rather than inferred from it. Our own results predict the ceiling will reproduce; the discriminating quantities are the *differences* between those front-ends and geometries, not the absolute number. Second, **port the winning chain on-chip**: the monostatic delay-domain detector fits an ESP32-S3 with roughly two orders of magnitude of headroom (Sec. III-D), which would make the sensor self-contained rather than tethered to an offline host. Third, and most open, **a localization front-end that does not lean on odometry**: the robustness reported in Sec. IX-E is real but inherited, and a WiFi measurement model able to carry absolute position on its own — rather than refine a motion estimate — would change what the modality can claim.

## XIII. CONCLUSION

Ambient WiFi is a practical, inexpensive localization sensor for automotive SLAM and a poor mapping sensor — and the reason it maps poorly is now specific and quantified. Most realistic-CSI detections are phantoms, and a several-metre range bias corrupts the rest; this is a property of the superresolution front-end and the bistatic geometry, not of the carrier, the bandwidth, or WiFi as such, as a controlled comparison against LiDAR and a 77 GHz radar on one substrate shows. The open problem for WiFi mapping is therefore the front-end, not the physics, and the clearest escape is a monostatic on-vehicle geometry, which nearly eliminates the phantoms and needs no known transmitter positions. We have shown the delay-domain reading holds on \$30 of real ESP32 silicon; measuring the phantom rate itself on that hardware,



and porting the winning chain on-chip, are the immediate next steps.

## REPRODUCIBILITY

All code, configs, and result data are in the repository above (AGPL-3.0); the simulation is Sionna RT; the hardware pipeline and firmware are included.

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